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Participatory Design & End-User Development

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Participatory Design & End-User Development: Building Bridges

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Abstract

Empowering end-users to be actively involved in the design, development and implementation of systems is a shared goal of the participatory design and end-user development communities. Yet, both communities have developed largely separately, both building upon their own specific set of knowledge, methods and practices. This workshop aims to identify common goals and a shared research agenda by bringing together researchers from both communities and stimulating the exchange of knowledge and the generation of new ideas.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; **HCI theory, concepts and models**; **Collaborative content creation**.

Keywords

End-user development, Participatory Design, Research agenda

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1 Workshop overview

1.1 Background

Participatory Design (PD) approaches have been proposed in Human-Computer Interaction (HCI) to foster the creation of interactive systems that are usable, acceptable, and meet users' expectations. This idea of PD is based on involving representative end users as active members of the design team [2, 3, 10]. Particularly, end users may directly participate in the creation of personas, scenarios [9], and low-fidelity prototypes [2], thus becoming co-designers of their systems.

However, an important hurdle that participatory design is not able to address effectively, is the arising, at the use time, of new system requirements that require the development of additional functionalities to support users' activities. In fact, it has been observed that systems need to evolve at use time to account for changing tasks, to deal with the variety of users, contexts, and needs, and to consider the introduction of new technologies. In this situation, meta-design has been proposed to design open systems than can evolve in the hand of the users [4–6], through methods and techniques that have been characterized as End-User Development (EUD) [7, 8]. Particularly, EUD empowers end users to create, modify, and extend digital artifacts throughout their entire life cycle, to fit users' evolving activities and needs [1].

While PD and EUD are two separate approaches, they are strongly related and complementary in their aims to involve end users in software development and HCI design.

1.2 Aims & Objectives

PD and EUD communities have evolved over many years with own conference series and rarely (if at all) meet to discuss common ideas. Yet, PD and EUD are united in their aims to empower end users of systems through (meta-)design and collaboration between developers and users. The aim of this workshop, therefore, is to foster collaboration and exchange of ideas between the EUD and PD communities.

Through this workshop, we aim to identify common goals and challenges in involving end users in design and development activities throughout the life-cycle of software applications. We explore in what ways the two fields are related, in what ways they are different, and how they can learn from each other to increase researchers' participation. Finally, we develop a shared research agenda for future collaboration and advancements in the field.

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2 Workshop planning

We are planning for a 1-day workshop to foster collaboration and exchange of ideas between the end-user development and participatory design communities. The activities planned for the workshop focus on interactive exchange of ideas, rather than static presentations. A draft program is outlined below:

- **09.00 – 09.30: Registration and welcome.**
- **09.30 – 10.00: Icebreaker & introductions.** Participants are asked to briefly introduce themselves, their research interests and their expectations for the workshop.
- **10:00 – 10:45: Short presentation of position papers.** Authors of accepted position papers are invited to briefly present their perspectives as an important step in creating a common understanding of commonalities and differences between EUD and PD approaches, methods, and practices.
- **10.45-11.15: Coffee break**
- **11.15-12.00: Continued position papers presentations**
- **12.00-13.00: Lunch**
- **13.00-14.30: Mapping the fields** Participants are invited to map their own research within PD and/or EUD on multiple dimensions (e.g., application domain, user characteristics, system characteristics, methods, trends, etc.), to characterize and visualize overlapping, complementary, and/or related aspects of their research.
- **14.30-15.00: Coffee break**
- **15.00-16.30: Collaborative charting of a common research agenda** After mapping the fields of PD and EUD, in this stage the participants are invited to split up in smaller groups to identify opportunities for collaboration, interaction and/or exploration based on the mappings they generated. Finally, the output of the subgroups is collected and structured into an agenda for future research that unites both fields.
- **16.30-17.00: Closing**

3 Promotional strategy

A CfP will be distributed on the main HCI mailing lists and social media. A website to advertise the workshop and to be used as repository of the accepted papers will be developed and maintained online after the workshop. Potential participants submit a six-page position paper, directly or indirectly addressing the topic and goals of this workshop. Everyone is asked to include a description of their work and its relation to the workshop theme. Upon acceptance, papers will be posted on the workshop website and attendees invited to read them prior to the workshop. Participants will be also encouraged to bring along examples of artifacts, systems and/or case studies relating to the workshop topic.

We would like to invite both researchers whose core research focus is on methodological work in PD or EUD, as well as researchers working on domain-related challenges (e.g., in healthcare, IoT, education, etc.) who apply PD or EUD approaches in their work.

A draft call for participation can be found here: https://docs.google.com/document/d/1Iru2XkI-Y7VNlRDIj7VqSrkFLTlExn6HKV-QwxV8JJZ4/edit?usp=sharing

4 Organizers' background

Daniel Tetteroo is assistant professor at the Department of Industrial Design at Eindhoven University of Technology. His research focuses enabling the design and development of personalized healthcare systems. Specifically, he is interested in empowering end-users to take control over technology in personally meaningful contexts. Daniel is a member of the End-User Development Symposium (IS-EUD) steering committee and has previously organized a successful CHI workshop on end-user development.

Sebastian Feger is a full professor at TH Rosenheim, Germany, and a researcher associated with LMU Munich, Germany. He aims to empower end users across various application domains through his research. Exemplary research threads include communicating smart home privacy considerations to diverse homeowners and bystanders and returning effective control over these appliances, as well as enabling the participation of the wider society in 3D design, modeling, and fabrication.

Daniela Fogli is full professor at the Department of Information Engineering, University of Brescia, Italy. Her research interests include methods for designing complex interactive systems, meta-design, end-user development, web usability and accessibility, collaborative robotics, and decision support systems. She is serving as chair of the steering committee of the International Symposium of End-User Development (IS-EUD) and is one of the organizers of the CoPDA Workshop Series "Cultures of Participation in the Digital Age".

Anders Mørch is professor at the Faculty of Educational Sciences, University of Oslo, Norway. His research is in the areas of computer-supported collaborative learning (CSCL), end-user development (EUD), and human-centered AI (HCAI). His current research is about didactic use of generative AI in compulsory education, using design-based research. He was program chair for NordiCHI-2006 and IS-EUD-2013 & 2019. He is one of the organizers of the CoPDA Workshop Series "Cultures of Participation in the Digital Age".

Antonio Piccinno is Associate Professor of Human-Computer Interaction (HCI) at the Computer Science Department of the University of Bari. His research interests focus on Human-Computer Interaction (HCI), specifically: Human-Centered Design (HCD), End-User Development (EUD), Interplay between Human-Computer Interaction and Software Engineering and, more recently, Secure software analysis and design. He is one of the organizers of the CoPDA Workshop Series "Cultures of Participation in the Digital Age".

Marie Utterberg Modén is a postdoctoral researcher at the Department of Applied Information Technology at University of Gothenburg. Her research interest revolves around theoretical, methodological and practical considerations around the empowerment of end users in digitalization of education. Currently, she works on developing new understanding of how to increase the agency of teachers concerning the design and use of AI-based systems in education to ensure fairness.

References

[1] Barbara Rita Barricelli, Fabio Cassano, Daniela Fogli, and Antonio Piccinno. 2019. End-user development, end-user programming and end-user software engineering: A systematic mapping study. *Journal of Systems and Software* 149

(2019), 101–137. https://doi.org/10.1016/j.jss.2018.11.041
[2] Susanne Bødker and Kaj Grønbæk. 1991. *Cooperative prototyping: users and designers in mutual activity.* Academic Press Ltd., London, UK, 331–358.
[3] Susanne Bødker and Ole Sejer Iversen. 2002. Staging a professional participatory design practice: moving PD beyond the initial fascination of user involvement. In *Proceedings of the 2nd Nordic conference on Human-computer interaction* (Aarhus, Denmark) *(NordiCHI '02)*. ACM, New York, NY, USA, 11–18.
[4] Gerhard Fischer, Daniela Fogli, and Antonio Piccinno. 2017. Revisiting and Broadening the Meta-Design Framework for End-User Development. In *New Perspectives in End-User Development*, Fabio Paternò and Volker Wulf (Eds.). Springer International Publishing, Cham, 61–97. https://doi.org/10.1007/978-3-319-60291-2_4
[5] Gerhard Fischer and Elisa Giaccardi. 2006. Meta-design: A Framework for the Future of End-User Development. In *End User Development*, Henry Lieberman, Fabio Paternò, and Volker Wulf (Eds.). Springer Netherlands, Dordrecht, 427–457. https://doi.org/10.1007/1-4020-5386-X_19
[6] Gerhard Fischer, Elisa Giaccardi, Yunwen Ye, Alistair G. Sutcliffe, and Nikolay Mehandjiev. 2004. Meta-Design: A Manifesto for End-User Development. *Commun. ACM* 47, 9 (sep 2004), 33–37. https://doi.org/10.1145/1015864.1015884
[7] Henry Lieberman, Fabio Paternò, and Volker Wulf (Eds.). 2006. *End User Development.* Springer, New York, NY. https://doi.org/10.1007/1-4020-5386-X
[8] Fabio Paternò and Volker Wulf (Eds.). 2017. *New Perspectives in End-User Development.* Springer International Publishing, New York, NY. https://doi.org/10.1007/978-3-319-60291-2
[9] M.B. Rosson and J.M. Carroll. 2002. *Usability Engineering: Scenario-Based Development of Human-Computer Interaction.* Morgan Kaufmann, San Francisco.
[10] Douglas Schuler and Aki Namioka (Eds.). 1993. *Participatory Design: Principles and Practices.* Erlbaum Associates Inc., Hillsdale, NJ, USA.